



Zoe Sleep Retreat

10 Interactive Experiments

Half Moon Bay Longevity Retreat | 2-3 Day Protocol



Overview

Run live, ad hoc experiments with 30 participants during the retreat without requiring pre-existing Zoe journey data. Each experiment uses group-based assignments delivered via the app to test physiological and lifestyle interventions.

Key Constraints

- ✓ 2-3 days maximum (not 10 days)
- ✓ Minimal pre-retreat data dependency
- ✓ Group-based (A/B/C assignments)
- ✓ App-delivered protocols
- ✓ Immediate measurable outcomes

1



The Temperature Sweet Spot

Test optimal sleep temperature by assigning participants to different room temperature zones

Group Assignment (30 Participants)

GROUP A

n=10

COOL

66°F

GROUP B

n=10

WARM

70°F

GROUP C

n=10

HOT

74°F



Experimental Timeline

Day 1

3:00 PM

Martin assigns groups via app

Day 1

10:00 PM

Evening check-in: Confirm room temp set correctly

Night 1

Sleep at assigned temperature

Day 2
7:30 AM

Complete sleep log: quality, latency, awakenings, thermal comfort

Day 2
5:00 PM

Results presentation + SWITCH: All groups try optimal 66-68°F

Day 3
7:30 AM

Complete sleep log + view individual improvement

Expected Outcomes

Sleep Quality (1-10 scale)

Group A (66°F)



Group B (70°F)



Group C (74°F)



Key Insights Delivered

✓ Personal optimal: "You sleep best at 66-68°F"

- ✓ Quantified impact: "Switching from 74°F to 67°F improved your sleep quality by 1.6 points"
- ✓ Actionable takeaway: "Set your home thermostat to 66-68°F for better sleep"

✓ Martin's Preparation Checklist

- ☐ Coordinate with Ritz-Carlton: Ensure all rooms have individual climate control
- ☐ Create group assignments in app (randomized, stratified by age/gender)
- ☐ Day 1, 3:00 PM: Assign groups via app
- ☐ Day 2, 8:00 AM: Pull sleep log data and generate charts
- ☐ Day 2, 5:00 PM: Present findings with visualizations

HIGH PRIORITY

Minimal Development

2



Morning Light Exposure Race

Test impact of morning light timing on energy and mood

Group Assignment (30 Participants)

GROUP A

n=10

EARLY
LIGHT

7:00-7:30

AM

30min walk

GROUP B

n=10

LATE LIGHT

9:00-9:30

AM

30min walk

GROUP C

n=10

CONTROL

Indoor
morning

No
intervention



Experimental Timeline

Day 1

6:00 PM

Martin assigns groups + sends morning walk assignments

Day 2

7:00 AM

Group A: Guided beach walk (30 min)

Day 2
9:00 AM

Group B: Guided beach walk (30 min)

Day 2
10:00 AM

ALL GROUPS: Complete morning check-in (energy, mood, focus)

Day 3
9:00 AM

Results: Light exposure → next-day energy correlation

Expected Outcomes

Morning Energy Levels (1-6 scale, measured at 10 AM)

Group A (7 AM light)



Group B (9 AM light)



Group C (No exposure)



Key Insights Delivered

- ✓ Optimal window: "Getting outdoor light before 8 AM boosted your energy by 35%"

- ✓ Circadian impact: "Early light improved your next-night sleep quality by 0.8 points"
- ✓ Actionable takeaway: "Aim for 30 minutes of outdoor light within 2 hours of waking"

MEDIUM PRIORITY

3



Caffeine Cutoff Challenge

Find your personal caffeine half-life and optimal cutoff time

Group Assignment (30 Participants)

GROUP A

n=10

STRICT
No caffeine
after 12 PM

GROUP B

n=10

MODERATE
No caffeine
after 2 PM

GROUP C

n=10

CONTROL
Normal
caffeine
intake



Expected Outcomes

Sleep Latency Improvement (Minutes to Fall Asleep)

Group A (12 PM cutoff) - Night 1: 18min → Night 2: 12min

6 min ✓

Group B (2 PM cutoff) - Night 1: 17min → Night 2: 14min**-3 min****Group C (Control) - Night 1: 18min → Night 2: 18min****0 min**

💡 Key Insights Delivered

- ✓ Personal sensitivity: "Your sleep latency improved by 6 minutes with a 12 PM cutoff"
- ✓ Half-life calculation: "At age XX, your caffeine half-life is ~6.5 hours"
- ✓ Optimal window: "Stop caffeine by 12 PM for best sleep (target <15% in system at bedtime)"

HIGH PRIORITY**Minimal Development**



Experiment Summary

#	Experiment	Duration	Primary Outcome	Expected Effect	Priority
1	 Temperature	2 nights	Sleep quality	+1.5 points (cool)	HIGH
2	 Morning Light	1 day + night	Energy, sleep	+1.2 energy (7AM)	MEDIUM
3	 Caffeine Cutoff	2 days	Sleep latency	-6 min (12PM cutoff)	HIGH
4	 Wind-Down Protocol	1 night	Sleep quality	+1.8 quality (full)	MEDIUM
5	 Meal Timing	2 nights	Digestive disruption	60% disruption (late)	MEDIUM
6	 Nap Study	1 day + night	Night sleep reduction	-40 min (45min nap)	LOWER
7	 Blue Light Blocking	1 night	Sleepiness onset	-55 min onset (blockers)	HIGH
8	 Gratitude Journal	1 night	Rumination, mood	-1.5 rumination	MEDIUM

#	Experiment	Duration	Primary Outcome	Expected Effect	Priority
9	 Breathing Exercise	1 night	Sleep latency	-4 min latency	LOWER
10	 Alcohol & Sleep	2 nights	REM sleep, awakenings	-25% REM	HIGH

Recommended Experiment Selection

Top 5 Experiments (Run These)

1. 🌡️ **Temperature Sweet Spot** - Easiest logistics, immediate impact, everyone participates
2. ☕ **Caffeine Cutoff** - Highly relatable, clear results, minimal development
3. 🔵 **Blue Light Blocking** - Just buy glasses, powerful results, "wow" factor
4. 🍷 **Alcohol & Sleep** - Very relevant to this demographic, eye-opening REM data
5. ☀️ **Morning Light** - Beautiful Half Moon Bay setting, social activity, clear benefits

Ready to Transform Sleep Science

These experiments will demonstrate Zoe's unique value: personalized, data-driven sleep optimization that delivers measurable results in just 2-3 days.